Teleoperation System Tutorial

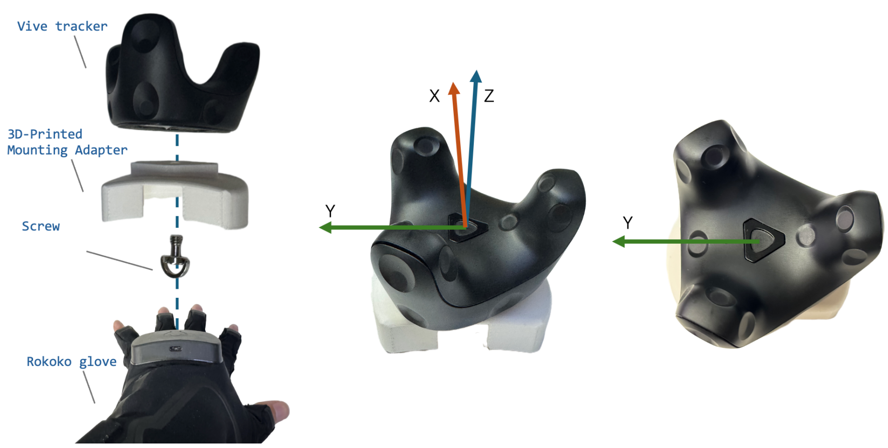
1 Hardware Setup

1.1 Bill of Materials

No.	Component Name	Quantity	Purchase Link
1	Rokoko Smart Gloves	1 Pair	https://www.rokoko.com/products/smartgloves
2	Power Bank (Supports 5V / 2A Output)	1	https://item.jd.com/100137426558.html
3	Wireless WiFi Router (Supports 5 GHz)	1	https://item.jd.com/100100368561.html
4	HTC Vive Tracker	2	https://www.vive.com/cn/
5	HTC Base Station	2	https://www.vive.com/cn/
6	Camera Stand	2	https://detail.tmall.com/item.htm
7	3D-Printed Mounting Adapter	2	The connector can be directly fabricated using a Bambu Lab 3D printer.
8	1/4-20 Camera Mount Screw TF3010	2	https://item.taobao.com/item.htm?id=709756999019

1.2 Assembly Guide

Assemble the hand motion capture system as shown in the figure below. Then, place the two HTC Vive Base Stations diagonally facing each other, as illustrated, to minimize occlusion during tracking. It is important to ensure that the Vive tracker is mounted to the glove with the fixed pose shown in the figure, as the system assumes that the orientation of the tracker coordinate frame is aligned with the simulator world coordinate frame.



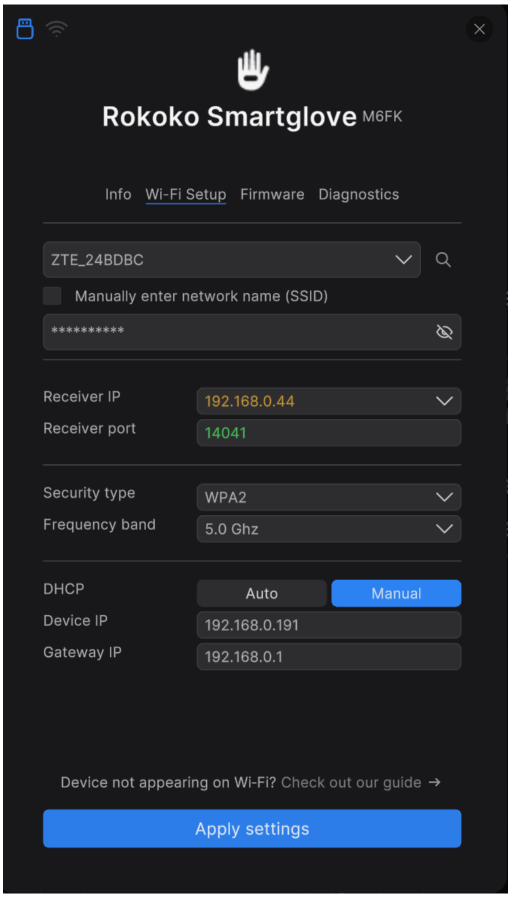
(a) motion capture system assembly

(b) overview

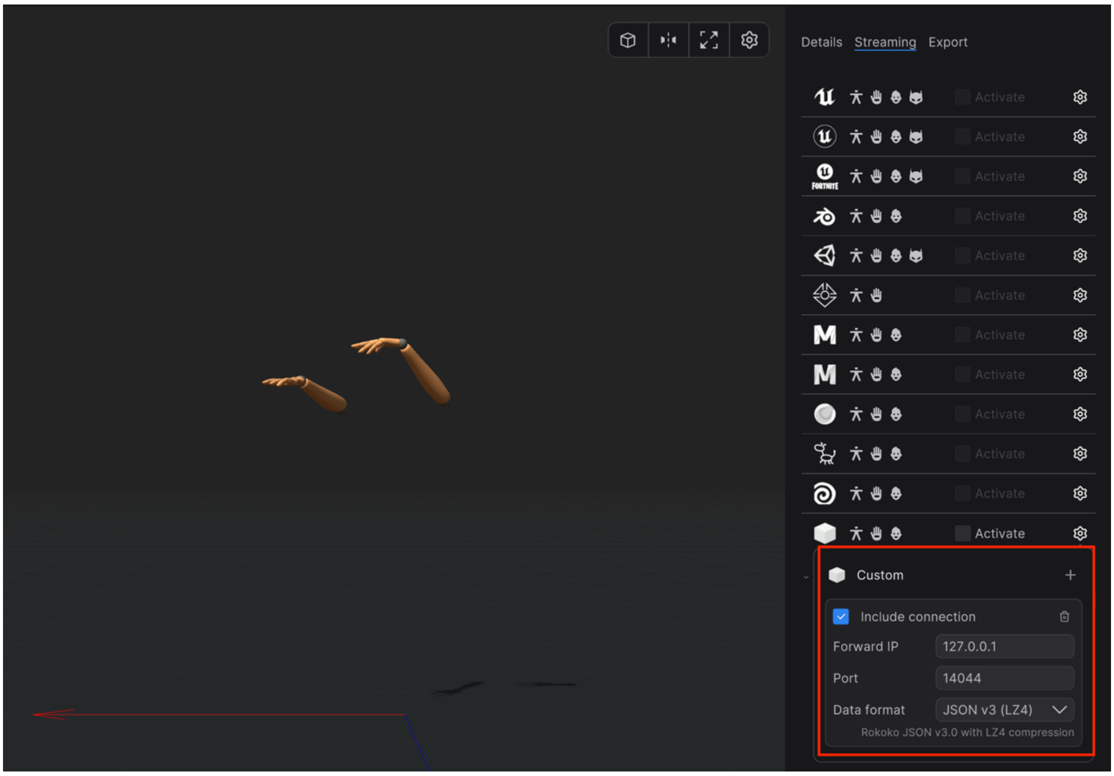
2 Software Setup

2.1 Rokoko Studio

First, download Rokoko Studio. After that, connect the gloves to the computer using a USB cable. The left and right gloves need to be configured separately. Connect the computer to the mobile WiFi network and ensure that the WiFi is set to the 5 GHz frequency band. Then, enter the computer's IP address into the "Receiver Device" field.



After the gloves are successfully connected to Rokoko Studio, create a new scene and navigate to the “Streaming” panel on the right side. Set the target port number and record it for later use. Then, set the data format to “JSON v3 LZ4”.



After the gloves are successfully connected to Rokoko Studio, create a new scene and navigate to the “Streaming” panel on the right side. Set the target port number and record it for later use. Then, set the data format to “JSON v3 LZ4” and click “Activate”.

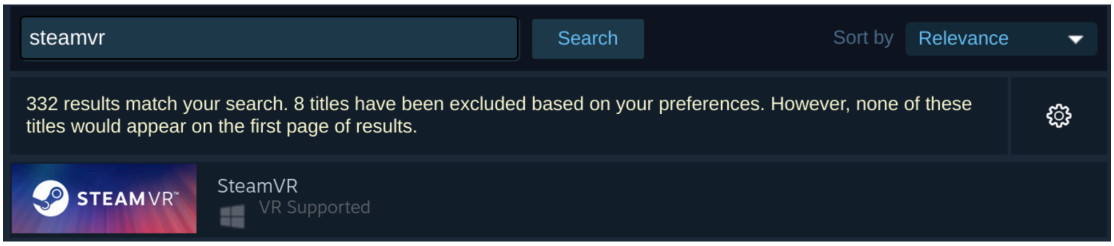
The glove data can now be streamed from Rokoko Studio to a specified port on the local computer. At this stage, the data can be used to collect training data for the retargeting neural network. During deployment, the streaming data should be forwarded to the server running the neural network. The network then outputs Allegro Hand joint angles, which are sent to the simulator to perform teleoperation of the robotic hand.

2.1 SteamVR

You may also choose to install SteamVR on the simulator machine (Ubuntu):

```
wget https://cdn.cloudflare.steamstatic.com/client/installer/steam.deb
sudo dpkg -i steam.deb
sudo apt-get update
sudo apt upgrade
```

After logging in with a Steam account, install SteamVR through the Steam client (Store → Search for "SteamVR").

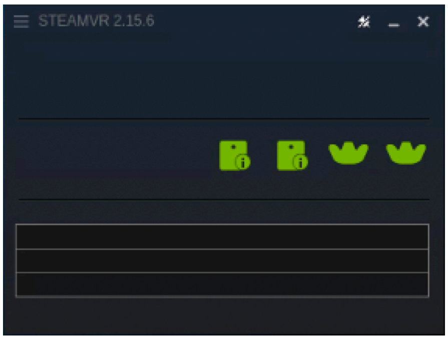


Connect the dongle to the computer, then power on the Vive Trackers and Base Stations. After that, press the button on the back of each Base Station and set the two Base Stations to “B” mode and “C” mode respectively.

Remove the HMD requirement: Since no Vive HMD (Head Mounted Display) is used in our setup, you need to modify the SteamVR configuration file. The default `default.vrsettings` file can be found at:

```
vim
home/yourDirectory/.local/share/Steam/steamapps/common/SteamVR/resources/settings/default.vrsettings
```

```
{
    "!!__WARNING__": {
        "000": "DO NOT EDIT THIS FILE TO CHANGE YOUR PERSONAL SETTINGS.",
        "001": "THIS FILE WILL BE REPLACED WHEN STEAMVR UPDATES.",
        "002": "These are the default values for settings which are not mentioned",
        "003": "in the user's personal steamvr.vrsettings file (use vrpathreg.exe",
        "004": "to find the location of this file). User settings which match the",
        "005": "default in this file are not written to the user settings file, but",
        "006": "any setting below may be placed in steamvr.vrsettings to override",
        "007": "these defaults."
    },
    "steamvr": {
        "contrast": 0.0,
        "requireHmd": false,
        "forcedDriver": "",
        "forcedHmd": "",
```



SteamVR can now successfully receive pose information from the Vive trackers.